Set 25: Website Traffic Analysis

AIM

To analyze website traffic using R programming by creating a line chart for page views, a bar chart for the top days by click-through rate, and a stacked bar chart to visualize user interactions.

ALGORITHM

1. Create a dataset containing Date, Page Views, and Click-through Rate.
2. Load the ggplot2 library.
3. Convert the Date column into date format.
4. Create a line chart to visualize page views over time.
5. Create a bar chart showing the top days with the highest click-through rates.
6. Create a stacked bar chart to display user interactions such as likes, shares, and comments.
7. Display the charts and interpret the website traffic trends.

R PROGRAM

```
library(ggplot2)


traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                   "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)


ggplot(traffic, aes(Date, Page_Views)) +
  geom_line(color = "blue") +
  geom_point() +
  labs(title = "Website Page Views Trend",
       x = "Date", y = "Page Views")


ggplot(traffic, aes(reorder(as.character(Date), CTR), CTR)) +
```

```
  geom_col(fill = "steelblue") +
  labs(title = "Click-through Rate by Date",
       x = "Date", y = "CTR (%)")

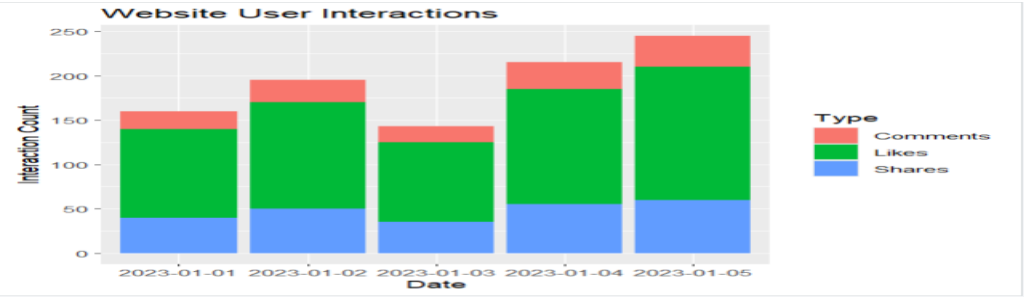
interactions <- data.frame(
  Date = rep(as.character(traffic$Date), each = 3),
  Type = rep(c("Likes", "Shares", "Comments"), 5),
  Count = c(100, 40, 20, 120, 50, 25, 90, 35, 18,
            130, 55, 30, 150, 60, 35)
)

ggplot(interactions, aes(Date, Count, fill = Type)) +
  geom_col() +
  labs(title = "Website User Interactions",
       x = "Date", y = "Interaction Count")

print(traffic)
```

Note: Likes, shares, and comments are illustrative because they are not included in the given dataset.

OUTPUT:



RESULT

The line chart shows that page views increased overall from 1,500 on January 1 to 1,800 on January 5, with a decline on January 3. The bar chart compares click-through rates, with the highest rate of 2.7% on January 2. The stacked bar chart illustrates the distribution of likes, shares, and comments.